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Sec: CSE 5-D

Lab 1: Requirements Engineering & UML Use-Case Modelling

Peer Skill Exchange & Mentorship Network

Task 1: Requirements Table

Req ID	Type	Description ("The system shall...")	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Learner Member to search for a Skill Mentor based on specific skill keywords.	High	When a Learner enters a keyword (e.g., "Python"), a list of mentors with that skill is displayed.	Core functionality for finding mentorship.
FR-002	Functional	The system shall allow a Learner Member to request a mentorship session with a selected Skill Mentor.	High	The Learner can select an available time slot and click "Request," changing the status to "Pending."	Required to initiate the skill exchange.
FR-003	Functional	The system shall allow a Skill Mentor to accept or decline a pending	High	When a Mentor clicks "Accept," the session status updates to "Scheduled"	Gives mentors control over their time.

Req ID	Type	Description ("The system shall...")	Priority	Acceptance Criteria	Rationale
		session request.		and notifies the Learner.	
FR-004	Functional	The system shall allow users to send direct messages to each other regarding a scheduled session.	Medium	A message typed into the chat interface appears on the recipient's screen within 2 seconds.	Facilitates pre-session communication.
FR-005	Functional	The system shall allow a Learner Member to leave a rating (1-5 stars) after a session is completed.	Medium	After completion, a feedback form appears; submitting it updates the Mentor's average rating.	Maintains quality and community trust.
NFR-001	Nonfunc.	The system shall return search results for Skill Mentors in under 2 seconds.	High	A timing log shows < 2 seconds from clicking "Search" to the list populating.	Ensures a responsive user experience.
NFR-002	Nonfunc.	The system shall encrypt all direct messages between members end-to-end.	High	Security audits confirm message payloads cannot be read in plain text in the database.	Protects user privacy.

Actors:

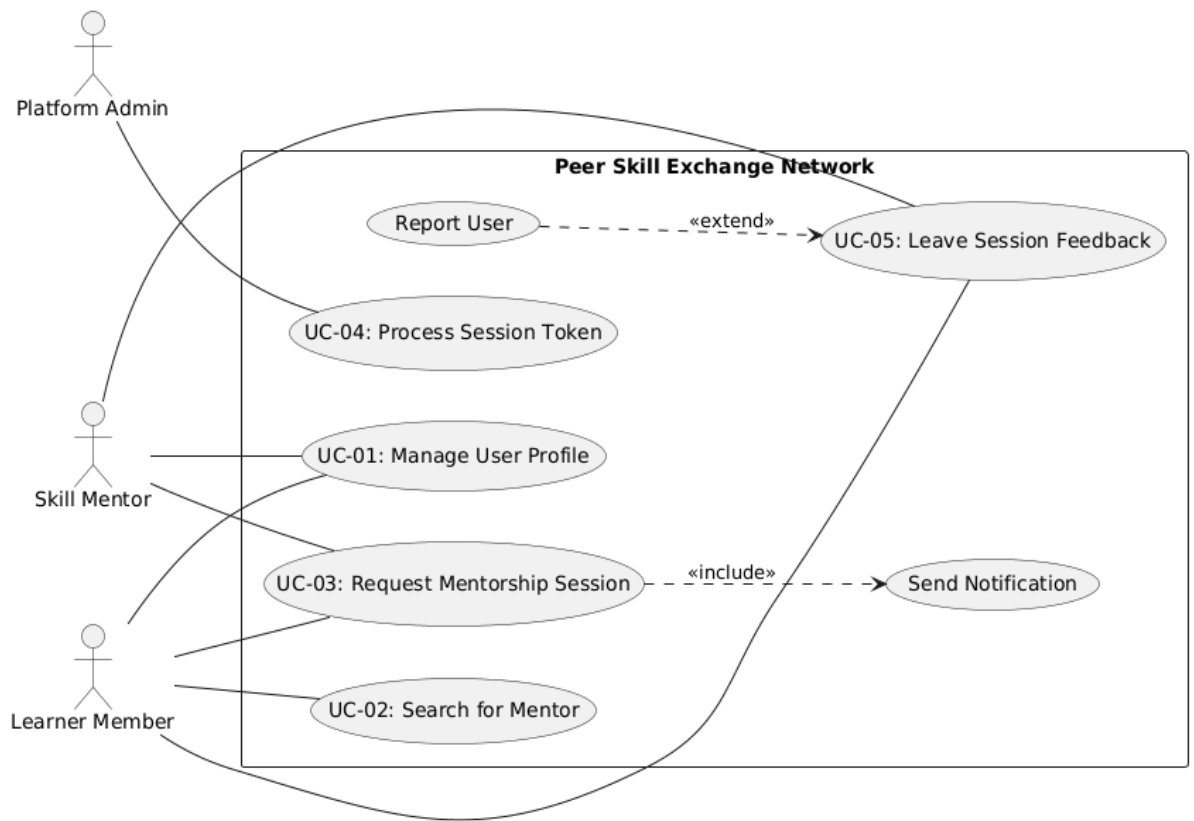
- Learner Member
- Skill Mentor
- Platform Administrator

Use Cases:

- UC-01: Manage User Profile
- UC-02: Search for Mentor
- UC-03: Request Mentorship Session
- UC-04: Process Session Payment/Token
- UC-05: Leave Session Feedback

Task2: UML Use-Case Diagram

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Task 3: Use-Case Flow

Use Case: Request Mentorship Session (UC-03)

Actors: Learner Member, Skill Mentor

Main Success Scenario:

1. Learner Member navigates to a Skill Mentor's profile page.
2. Learner Member clicks the "Request Session" button.
3. The system displays a calendar with the Mentor's available time slots.
4. Learner Member selects a preferred date and time.
5. Learner Member types a brief description of what they want to learn and submits the request.
6. The system sets the request status to "Pending" and sends a notification to the Skill Mentor.
7. The Skill Mentor reviews the request and clicks "Accept."

8. The system updates the status to "Scheduled," blocks off the time on both users' calendars, and sends a confirmation to the Learner Member.
9. Use case ends successfully.

Alternate Flow (7a. Mentor Declines Request):

- 7a1. The Skill Mentor reviews the request and clicks "Decline."
- 7a2. The system prompts the Mentor to optionally provide a reason.
- 7a3. The system sets the status to "Declined" and notifies the Learner Member.
- 7a4. The system prompts the Learner Member to request a different time or search for another Mentor.
- Use case ends.